Yi Hu

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Personal Summary

PhD researcher specialized in **embodied AI** and **surgical robotics**. Developed **Vision-Language-Action (VLA) model** and **Reinforcement Learning (RL)/Imitation Learning (IL)** frameworks for Franka and da Vinci surgical robots. Strong background in SLAM, kinematics/dynamics, and control theory.

Education

PhD in Biomedical Engineer — Jan. 2021 – Exp. Apr. 2026
University of Alberta — *Edmonton, Canada*

MSc in Control Theory and Control Engineering — Sept. 2017 – Jun. 2020
Sichuan University — *Chengdu, China*

B.S. in Process Equipment and Control Engineering — Sept. 2013 – Jun. 2017
Hefei University of Technology — *Hefei, China*

Project Experience

Human-Centered Autonomy Lab — Jan. 2024 – Present
Graduate Researcher, University of Alberta — *Edmonton, Canada*

- Designed a **VLA** policy integrating **3D point cloud reconstruction** (PointNet) and VLM feature projection, fused via a **Mixture-of-Experts** (MoE) action head to condition robot joints; trained on the LIBERO-Spatial environment (10 reasoning tasks with 500+ demonstrations), achieving >**96**% success rate.
- Built an end-to-end **VLA** training and deployment pipeline: collected and post-trained a **Pi0.5 model** on >300GB multimodal robot data across 5 manipulation tasks with **750+ demonstrations**, and deployed real-time inference on a real-world Franka robot, achieving >**95**% success rate.
- Engineered a real-time motion policy for **dynamic fast manipulation** by replacing multi-step diffusion sampling with a single-step **diffusion model**; trained on 100 demonstrations, achieving **7–10** inference latency reduction and validated on a real Franka robot ball-catching task.
- Developed a RL framework that transfers from large-scale human activity datasets (**32+ tool-use tasks**) to robotic surgical subtasks, enabling precise patient-side manipulation and endoscopic camera control with **13.5** improved sample efficiency; validated via sim2real transfer from PyBullet to a real-world da Vinci surgical robot.

Telerobotic and Biorobotic Systems Group — Jan. 2021 – Dec. 2023
Graduate Researcher, University of Alberta — *Edmonton, Canada*

- Built an end-to-end autonomous ultrasound scanning system on a Franka robot, integrating **classic control** using an Axia80-M20 force–torque sensor with **UNet–based** deep learning image processing via DVI2USB video capture, achieving >**80**% **accuracy** at >**10 Hz** closed-loop control.
- Developed an autonomous endoscopic camera control system for the da Vinci surgical robot operating on real-world noisy surgical data, with automatic **data processing** of unsafe or failed actions and a safety-critical, collision-aware **motion planning** framework.
- Developed a **real-time teleoperation** pipeline that transformed **free-hand movements** into **6-DoF robot** control; Integrated a lightweight **gesture classifier** trigger discrete robot actions.

Perception, Motion Control and Intelligent Robot Innovation Lab — Sept. 2017 – Dec. 2020
Graduate Researcher, Sichuan University — *Chengdu, China*

- Implemented a real-time **event-camera SLAM** pipeline for **mobile robot navigation** with **scene understanding**, enabling 6-DoF **motion tracking** and **3D reconstruction** through optimization-based mapping and keyframe back-end refinement. Built a **physics-based modeling** in Unity for mobile robot navigation.
- Designed robust **nonlinear control** algorithms for **joint-space control**, accounting for **actuator saturation** and incorporated **disturbance observers**, with **system stability** guaranteed through Lyapunov analysis.
- Derived **kinematic and dynamic models** for a wheel mobile robot, enabling motion planning, control design, and trajectory tracking.

Skills

Programming: Python, MATLAB/Simulink, C++
Robotics: ROS, ROS2, MuJoCo, PyBullet, Isaac Gym, Gazebo, RViz
ML & AI: PyTorch, TensorFlow, Diffusion Models, Transformers, RL, IL
Systems & Tools: Linux, Git, CUDA, SolidWorks, COMSOL, AutoCAD